7CS039 Statistics for AI & Data Science

Week 5 Applications I

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1 Introduction

Last time we looked at Z-tests and t-tests. These are examples of what are called parametric tests. Parametric tests are those that make assumptions about the parameters of the population distribution from which the sample is drawn. This is often the assumption that the population data are normally distributed. Non-parametric tests are "distribution-free" and, as such, can be used for non-Normal variables. The table below shows the non-parametric equivalent of a number of parametric tests.

Parametric Test	Non-Parametric Test
Paired t-test	Wilcoxon Rank Sum Test
Unpaired t-test	Mann-Whitney U Test
Pearson Correlation	Spearman Correlation
One way Analysis of Variance	Kruskal Wallis Test

Given some data, a non-parametric test in general has lower *power* - i.e. is less likely to find a "significant" difference, when a real difference exists - than the corresponding parametric test. But "significance" based on false assumptions is a serious error.

2 The Mann-Whitney U-Test

Two non-parametric tests based on "rank sums" were proposed in the 1940s for comparing two populations when there are indications that the conditions necessary for applying the independent samples t-test are not met. One test, proposed by F. Wilcoxon in 1945, is called the **Wilcoxon rank-sum test**, and the other, proposed by Mann and Whitney in 1947, is commonly referred to as the **Mann–Whitney U–test**. These two tests are in fact equivalent, and lead to identical conclusions, we will only look at the Mann-Whitney U test here.

Example

Consider the data presented in the table below which records the charitable contributions (in dollars) made during a year by two samples of households picked at random from two communities in the same economic bracket. We are interested in testing whether the average amounts contributed by the two communities may be equal.

Community A	Community B
120	42
155	190
88	320
420	680
360	82
60	130
650	500
82	32
180	160
98	82
	890
	3020

We proceed as follows:

H_0: both samples come from the same population. $[D_A = D_B]$

H_1: both samples do not come from the same population. $[D_A \neq D_B]$

i) Combine the two samples, and order them, starting with the least data value. Here I have underlined the data which come from the Community A sample.

$$32 \quad 42 \quad \underline{60} \quad 82 \quad 82 \quad \underline{82} \quad \underline{88} \quad \underline{98} \quad \underline{120} \quad 130 \quad \underline{155}$$

$$160 \quad \underline{180} \quad 190 \quad 320 \quad \underline{360} \quad \underline{420} \quad 500 \quad \underline{650} \quad 680 \quad 890 \quad 3020$$

ii) Assign rank 1 to the least data item, rank 2 to the second-least etc. Note that as the 4th, 5th and 6th ranks are tied, they are each given a rank of $\frac{4+5+6}{3} = 5$.

32	42	$\underline{60}$	82	82	$\underline{82}$	$\underline{88}$	$\underline{98}$	$\underline{120}$	130	$\underline{155}$
1	2	3	5	5	5	7	8	9	10	11
160	$\underline{180}$	190	320	$\underline{360}$	$\underline{420}$	500	$\underline{650}$	680	890	3020
12	13	14	15	16	17	18	19	20	21	22

iii) Find the *Rank Sums* of the two samples. i.e. Add up the ranks of the data that originates in sample A to find the rank sum R_A, and likewise find R_B. Here $R_A = 3 + 5 + 7 + 8 + 9 + 11 + 13 + 16 + 17 + 19 = 108$
and $R_B = 1 + 2 + 5 + 5 + 10 + 12 + 14 + 15 + 18 + 20 + 21 + 22 = 145$

Example ctd.

iv) Let the size of sample $A = n_1$ and the size of sample $B = n_2$. Note that if the size of the two samples combined is m, (i.e. $n_1 + n_2 = m$) then $R_A + R_B = 1 + 2 + 3 + \cdots + m = \frac{m(m+1)}{2}$, this is a useful check. In this case $m = 22$, and $R_A + R_B = 108 + 145 = 253$, and also $\frac{22\times 23}{2} = 253$. If these two values disagree, we know that we have gone wrong somewhere. Note also that a large value of R_1 implies a small value of R_2 and vice versa.

v) Let:

$$U_1 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 \tag{1}$$

$$U_2 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 \tag{2}$$

$$U = \text{the smaller of } U_1 \text{ and } U_2 \tag{3}$$

Here $n_1 = 10$, $n_2 = 12$, $R_1 = 108$, $R_2 = 145$. Thus

$$U_1 = (10)(12) + \frac{(12)(13)}{2} - 145 = 53 \tag{4}$$

$$U_2 = (10)(12) + \frac{(10)(11)}{2} - 108 = 67 \tag{5}$$

and hence $U = 53$. This is U_{obs}. The smaller U_1 is the larger U_2 is, or vice–versa. Thus if U is small, $|U_1 - U_2|$ is large. Hence the smaller U is, the greater $|U_1 - U_2|$ is, and the more likely it is that the populations from which samples A and B were selected differ.

vi) Look up the critical value of U, U_{crit} in `R`.

Here we see that for $n_1 = 10$ and $n_2 = 12$, $U_{\text{crit}} = 29$.

vii) We will reject H_0: The averages of the two samples are equal, if $U_{\text{obs}} < U_{\text{crit}}$.

Be aware that this is the reverse of the usual procedure. Here we are rejecting H_0 if U_{obs} is *less* than U_{crit}. This makes more sense if one considers the fact that a small value of U_{obs} indicates a large difference between U_1 and U_2.

In this case $U = 53 > 29 = U_{\text{crit}}$, so we do not reject H_0, (at $\alpha = 0.05$). There is not (significant) evidence that the distributions of the contributions from communities A and B differ.

This is the Mann–Whitney U–Test.

3 Chi-Square Tests

The χ^2 distribution is defined by: $Y = Y_0 \chi^{\nu-2} e^{-\frac{1}{2}\chi^2}$, but you will be glad to know that we will not be the least bit concerned with this theoretical distribution. The χ^2 distribution has many uses, for example, it may be used to perform tests of variances in a similar manner to which we use the Z and t–distributions to perform tests of means.

3.1 Calculating the Observed Test Statistic

For the χ^2 test for independence of two variables we calculate the observed test statistic as

$$\chi^2_{\text{obs}} = \sum \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}}$$

Expected counts are the values that the data would take if the null hypothesis was true. For example, say that the null hypothesis being tested is that the number of males and females in Wolverhampton is equal; then if a random sample of 200 people is taken, the expected counts would be 100 males and 100 females. If the null hypothesis was that the ratio of males to females in Wolverhampton is $4:6$ the expected values would be 80 males and 120 females.

3.2 χ^2–test for the independence of two variables

This test is also known as the χ^2–*test for contingency tables.*

Example

A contingency table is a table such as:

	A	B	C
Passed	24	56	33
Failed	11	40	24

Which illustrates the numbers of people that passed or failed driving tests administered by three different examiners A,B , and C. The obvious question to ask is: "Is the outcome of the test dependent (or contingent) on the examiner giving the test?" Thus:

H_0: The outcome of the driving test is independent of the examiner.

H_1: The outcome of the driving test depends upon the examiner.

Example ctd.

Let us look at the table of observed counts again:

	A	B	C	*Total*
Passed	24	56	33	113
Failed	11	40	24	75
Total	35	96	57	188

To continue with a χ^2–test we need to calculate the expected counts if H_0 is true, i.e. if "Outcome" and "examiner" are independent variables. We do this as follows: From the contingency table we see that a total of $113/188$ pass and a total of $35/188$ are examined by "A". i.e.

$$P\,(\text{Pass}) = \frac{113}{188} \qquad \text{and} \qquad P(A) = \frac{35}{188}$$

If X and Y are independent events then

$$P\,(X \text{ and } Y) = P\,(X) \times P\,(Y)$$

Hence here:

$$P\,(\text{Pass and } A) = P\,(\text{Pass}) \times P\,(A) = \frac{113}{188} \times \frac{35}{188} = \frac{3955}{35344}$$

This is the probability that any one individual selected at random would have both passed the driving test *and* have been examined by examiner "A". Thus as there are 188 people in the sample the expected count for "$Pass$ and A" is:

$$\frac{3955}{35344} \times 188 = 21.04$$

Similarly, if we wish to calculate the expected number of people that fail with examiner B, we see that $P(\text{fail}) = \frac{75}{188}$, $P(B) = \frac{96}{188}$, thus $P(\text{Fail and } B) = \frac{75}{188} \times \frac{96}{188} = \frac{450}{2209}$, and hence of the 188 people we would expect

$$\frac{450}{2209} \times 188 = 38.3$$

to fail with examiner B. Note the emerging pattern: in general the expected count in row i, column j is:

$$\frac{(\text{Total Row } i) \times (\text{Total Column } j)}{\text{Grand Total}}$$

For instance, the expected number to pass with examiner C is:

$$\frac{(\text{Total Pass}) \times (\text{Total C})}{\text{Grand Total}} = \frac{113 \times 57}{188} = 34.26$$

Example ctd.

Continuing in this fashion we obtain the following table of expected counts (rounded to one decimal place).

	A	*B*	*C*	*Total*
Passed	21.0	57.7	34.3	113.0
Failed	14.0	38.3	22.7	75.0
Total	35.0	96.0	57.0	188.0

We can then incorporate the observed/expected counts in one table.

	A	*B*	*C*	*Total*
Passed	24 (21.0)	56 (57.7)	33 (34.3)	113.0
Failed	11 (14.0)	40 (38.3)	24 (22.7)	75.0
Total	35.0	96.0	57.0	188.0

We can calculate χ^2_{obs}.

$$\begin{aligned}
\chi^2_{\text{obs}} &= \sum \frac{(\text{obs} - \text{exp})^2}{\text{exp}} \\
&= \frac{(24-21)^2}{21} + \frac{(56-57.7)^2}{57.7} + \frac{(33-34.3)^2}{34.3} \\
&+ \frac{(11-14)^2}{14} + \frac{(40-38.3)^2}{38.3} + \frac{(24-22.7)^2}{22.7} \\
&= 0.43 + 0.05 + 0.05 + 0.64 + 0.08 + 0.03 \\
&= 1.28
\end{aligned}$$

For a $r \times c$ contingency table, the number of degrees of freedom, ν, $= (r-1)\times(c-1)$. Thus here we have $(2-1)\times(3-1) = 2$ degrees of freedom. From `R` we obtain $\chi^2_{\text{crit}} = 5.99$ at $\alpha = 0.05$ with two degrees of freedom. Since $\chi^2_{\text{obs}} = 1.28 < 5.99 = \chi^2_{\text{crit}}$ we do not reject H_0. There is not significant evidence that there is a relationship between the outcome of the test and the examiner.

This is the χ^2 test for independence.

The two tests we've looked at today are quite calculation intensive. In practise we will carry out both in `R` but there are subtleties about the above two tests that require us to understand the calculations in more detail than perhaps we do with some other concepts.